Rohan Kumar

Final Year Undergraduate, IIT Kanpur
Major: Biological Sciences and Bioengineering
Minor: Industrial and Management Engineering

rohank23@iitk.ac.in
Rohan Kumar
bt_hai | rohan53532211

Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023-Present	**6.8**/10.0
Intermediate/+2	CBSE(XII)	Sarvodaya Bal Vidyalaya Rly Clny, New Delhi	2023	93.4%
Matriculation	CBSE(X)	GBSSS No.2 Rly Cly, New Delhi	2021	80.6%

Scholastic Achievements

- Achieved a maximum rating of **1499 on Codeforces**, showcasing strong algorithmic problem-solving.
- Achieved a **3-Star rating on CodeChef**, demonstrating strong competitive programming proficiency.
- Secured **Global Rank 28 in CodeChef Starters 200**, showcasing strong competitive programming skills.
- Cleared the **NISM-Series-VIII** certification examination conducted under **SEBI** regulations in **Equity Derivatives**.
- Secured **Top-10 at Strategy Storm, IIT Guwahati**, among **2,000+ participants**, through strategic analysis.

Work Experience

RamanIQ.ai *May'26 - Jul'26*
Software Developer Intern

- Built a **full-stack document platform** with **Next.js & FastAPI**, enabling structured document management.
- Designed a **modular service architecture** separating frontend, APIs, data processing, and persistence components.
- Modeled complex document hierarchies using **PostgreSQL & SQLAlchemy** with indexed and relational data structures.
- Developed reusable **REST APIs** powering document search, asset management, and seamless frontend-backend workflows.
- Automated **document ingestion workflows** for multi-format parsing, chunking, metadata extraction, and persistence.

Research Experience

IPO Price Dynamics in the Indian Market *May'26 - Aug'26*
Prof. Abhinava Triphati | Dept. of Management Sciences

- Studied **1,000+ Indian IPOs** through **underpricing, information-asymmetry, and price-discovery** frameworks.
- Reviewed **seminal IPO literature** and derived **empirical hypotheses** around demand, pricing, and investor behavior.
- Built an **IPO-level database** integrating price bands, **QIB/NII/RII demand**, anchors, promoter stakes, and listing data.
- Automated **DRHP/RHP extraction** using **Python, OCR, NLP, PyMuPDF, and pdfplumber** to structure disclosures.
- Built **data-cleaning and feature-engineering workflows** for **underpricing, returns, and investor demand** analysis.

Position of Responsibility

Head, Udghosh'26 *May'26 - Present*
Annual Sport Fest — IIT Kanpur

- Spearheaded an **85+ member, 4-tier security team**, managing campus-wide deployment and **3 concert nights**.
- Coordinated with **Gymkhana, SIS, administration, and district authorities**, managing access control for **50+ events**.

Coordinator, Ultimate Frisbee Club *May'25 - May'26*
Games And Sports Council — IIT Kanpur

- Transformed the Society into a Club, managing an **INR 80K budget** and overseeing club operations and representation.
- Led the Frisbee team as **Captain**, organizing **3 intra-college tournaments** and driving **60% participation growth**.
- Developed the Club's **official website and strategy simulation tool**, supporting tactical analysis and training.

Projects

Ultimate Frisbee Strategy Platform *Dec'25-Mar'26*
Self Project

- Built a platform using **React + Vite** to **create, simulate, save, discover, and share** interactive frisbee strategies.
- Designed **Node.js + Express REST APIs** supporting CRUD operations, search, pagination, and versioning.
- Engineered a **deterministic simulation engine** using **keyframes, LERP, Bézier curves, and parabolic motion**.
- Built **event-driven architecture** with a **state machine**, streamlining game-event validation and cutting latency by **15%**.
- Built **real-time collaboration** for concurrent strategy editing and sharing, with **integrated chat**, authentication, undo/redo, validation, and responsive UI.

Mess Automation System *Jan'26-Apr'26*
Software Development: Prof.Indranil Saha

- Architected a scalable **tri-service system** using **React.js, Node.js/Express, and FastAPI** for real-time automation.
- Engineered **facial recognition authentication** with **OpenCV LBPH** for real-time attendance and verification.
- Developed an automated **face training pipeline** for dataset generation, training, recognition, and **PostgreSQL sync**.
- Automated **backend workflows** for dues, rebates, meal bookings, and feedback using **REST APIs and PostgreSQL**.
- Built interactive **analytics dashboards** for attendance, extra-buying, and revenue metrics using **backend data**.

Memory-Efficient Versioned File Indexer *Mar'26 - Apr'26*
Software Development: Prof.Indranil Saha

- Implemented a **memory-efficient C++ indexer** using fixed-size buffers for scalable word-level processing of large files.
- Built an **STL unordered_map-based index** for fast word-frequency retrieval with optimized memory and file I/O.
- Designed a **modular OOPs query system** supporting word counts, version differences, and top-K frequency queries.

Interest Rate Curve & Swap Risk Engine *May'26-Jun'26*
Self Project

- Engineered a modular **C++17 pricing engine** using **OOPs and polymorphism** for reusable financial computations.
- Designed reusable **instrument, curve, and interpolation interfaces** with modular components for scalable extension.
- Built a robust **CSV data pipeline** for market-data ingestion, maturity parsing, curve construction, and result generation.
- Developed an analytical **sensitivity engine** using chain-rule gradient propagation for efficient risk computation and testing.

Skills

- **Languages:** C, C++, Java, Python, JavaScript, SQL
- **Frameworks:** React, Next.js, Node.js, FastAPI, REST APIs
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras
- **Data & Tools:** PostgreSQL, MySQL, MongoDB, Git, GitHub

Extracurricular Activities

- Secured **Second Runner-Up at Udghosh'25**, leading the institute **Ultimate Frisbee team as captain**.
- Secured **Bronze in Ultimate Frisbee** during Inferno'26 (Inter-Hall Tournament), leading the **Hall-3 team as captain**.

Relevant Courses

Software Development & Operations
Fundamentals of Data Science & ML
Fundamental of Computing
Security Analysis for Investments
Applied Probability & Statistics
Ordinary Differential Equations
Portfolio Management
Single Variable Calculus
Linear Algebra